

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Standard & Compound Units

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

STANDARD & COMPOUND UNITS · CH4 · L1

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The Map

This lesson collapses Standard & Compound Units into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Pressure: force over area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Density: mass and volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Speed: distance over time

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Volume unit conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 39 website questions, 39 checked solutions, 4 local PDFs read.

01 Pressure: force over area

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1H Q13

Calculate area

$$\text{pressure} = \frac{\text{force}}{\text{area}}$$

The cube rests on one

$$w^2 = 400$$

Finish: 8000 cm³.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

02 Density: mass and volume

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$\text{mass} = \text{density} \times \text{volume}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q9

Calculate volume

$$\text{mass} = \text{density} \times \text{volume}$$

Finish: 2895 g.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

03 Speed: distance over time

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$3 \text{ h } 36 \text{ min} = 3.6 \text{ h}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2HR Q5

The time is

$$3 \text{ h } 36 \text{ min} = 3.6 \text{ h}$$

Average speed

$$2470 \div 3.6 = 686.111 \dots$$

Finish: 686 km/h.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

04 Volume unit conversion

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Standard & Compound Units question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$1 \text{ m} = 100 \text{ cm}$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q2

Calculate value

$$1 \text{ m} = 100 \text{ cm}$$

Calculate value

$$1 \text{ m}^3 = 100^3 \text{ cm}^3 = 1,000,000 \text{ cm}^3$$

Finish: 32,400,000 cm³.

FORMULA BANK

$$\text{speed} = \frac{\text{distance}}{\text{time}}, \rho = \frac{m}{V}, \text{pressure} = \frac{F}{A}.$$

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Pressure: force over area
"work out", "show", "hence", a familiar exam phrase	Density: mass and volume
"work out", "show", "hence", a familiar exam phrase	Speed: distance over time
"work out", "show", "hence", a familiar exam phrase	Volume unit conversion

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the

trigger, write the first line, and let the working follow.